Vertex Hub: an AI-driven PLM project accelerator

V.E.R.T.E.X. AI, Value Extraction and Reconciliation for Technical Excellence. *The intersection of PLM ontologies and agile delivery.*

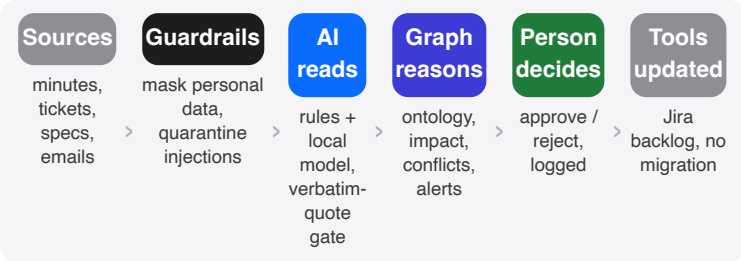
PROBLEM

- **Every PLM project starts from zero.** Each client has its own editor (3DEXPERIENCE, Windchill, Teamcenter), vocabulary (FR/EN, “Plant BOM” vs MBOM), tools and method. Setting up the framework takes weeks.
- **Decisions are scattered** across minutes, tickets, specs and emails. Nobody sees that a decision in one meeting breaks an interface agreed in another.
- **Inconsistencies surface in test, not in the meeting:** rework on interfaces, test cases and data migration.
- **Generic AI is not an option:** it invents facts and sends client data outside.

APPROACH AND METHODOLOGY

- **Built as a PLM project would be run:** CR/CO/CA change process, EBOM/MBOM/PBOM, ERP and MES interfaces, a change board that decides.
- **Realistic synthetic data:** an automotive client (ACME, 3DEXPERIENCE + SAP + MES) with minutes, workshops, status reports, backlog, tickets and tests, and a medtech client (Windchill, Scrum) to prove reuse.
- **Measured, not claimed:** a ground truth with a **holdout** the rules never saw; 46 automated tests; an injection benchmark.
- **Security first:** local model, masking and quarantine before any AI call, every action audited.

SOLUTION



1	**Framework from client data and method**	One sentence → client profile → backlog in the client's PLM vocabulary, structured for SAFe, Scrum or Waterfall
2	**Ontologies, bilingual glossary, updates**	15 FR/EN concepts, editor profiles for 3DEXPERIENCE, Windchill and Teamcenter; new terms proposed, validated by a steward
3	**Ingest and detect inconsistencies**	Minutes, tickets, specs, emails → facts with source quote → conflicts and dated alerts
4	**Connect without migration**	Reads the files teams already write; pushes decisions to Jira (mock in demo, cloud connector ready)
5	**Multi-criteria decision support**	Impact map, trade-offs per proposal, approve / reject with an audit trail
6	**No hallucination, no leak**	Quote must exist verbatim, IDs must exist; local model; personal data masked; injections quarantined
7	**Governance and billing**	Named owners and release gates; Build / Run / Support pricing (below)

RESULTS ON THE DEMO PROJECT

14 days
earlier warning: the 80→40 character truncation flagged on 21 Sept, not at test on 5 Oct

1 fixed, 2 new
ERP proposal analysed: fixes INT-01, but outdates TC-205 and breaks a naming rule; 5 stories, 4 tests impacted

35 / 1
personal-data items masked / injection quarantined before any AI call

7·15·27
epics, capabilities, features generated for a new medtech client in seconds

2 / 3 / 6
R2026x upgrade: new truncation risks / interfaces to retest / tests to rerun

0 / 18
false alarms of the injection filter on real project language (0% on benchmark)

8/8 → 0/4
rules alone: perfect on seen docs, blind on the holdout. Why the AI is needed

3/4
holdout facts found by rules + local model (qwen2.5:7b); local model alone: 3/4

GOVERNANCE

- The AI proposes, **a named person approves**; every decision is logged.
- Client-side ontology steward validates new terms; the integrator owns catalogue, editor profiles and connectors.
- Release gates: a new model or rule ships only if holdout recall holds and false alarms stay at 0.

BILLING MODEL (assumptions to validate)

Build, once per client	connectors, vocabulary, framework in their method	**~€15k** fixed
Run, per project	local model, ingestion, alerts, audit trail	**~€1.5k**/month
Support, yearly	new connectors, ontology, model upgrades re-measured	**~15%** of build

LIMITS WE STATE OPENLY

Synthetic data only · Jira connector tested as a mock · name detection relies on known names · injection filter tuned for zero false alarms, so it catches only obvious attacks (12% recall on the benchmark) · prices are working assumptions.

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Python, local LLM (Ollama), networkx, Streamlit